

Prescriptive process models are traditional software development models that follow a structured and well-defined sequence of steps. They prescribe a strict approach to software development, ensuring predictability, discipline, and organization in the development process. These models are useful for projects that require well-documented and sequential progress.

4.1 Types of Prescriptive process models:

- The Waterfall Model
- V model
- Incremental Process Models
- Evolutionary Process Models

4.1.1. The Waterfall Model:

A Waterfall model is the simplest model of software development paradigm. It says the all the phases of SDLC will function one after another in linear manner. That is, when the first phase is finished then only the second phase will start and so on.

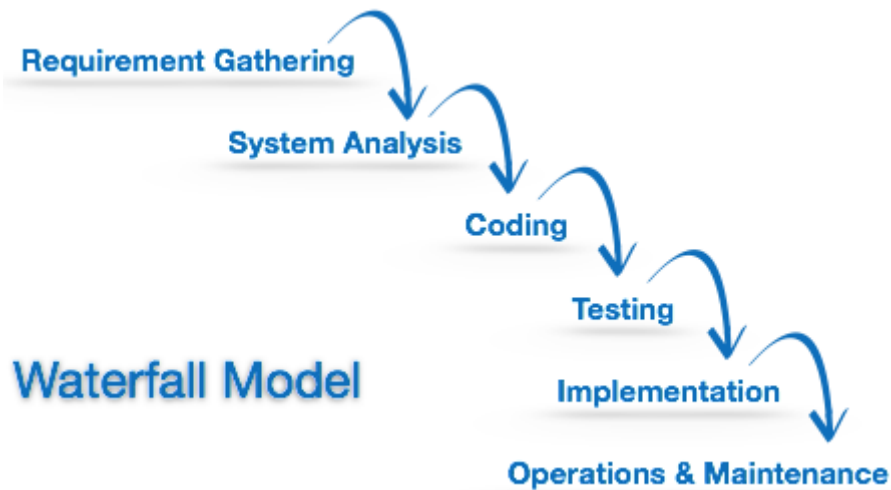


Fig 4.1

This model assumes that everything is carried out and taken place perfectly as planned in the previous stage and there is no need to think about the past issues that may arise in the next phase. This model does not work smoothly if there are some issues left at the previous step. The sequential nature of model does not allow us go back and undo or redo our actions.

This model is best suited when developers already have designed and developed similar software in the past and is aware of all its domains.

Advantages of waterfall model:

- This model works for small projects because the requirements are understood very well.
- The waterfall model is simple and easy to understand, implement, and use.
- All the requirements are known at the beginning of the project, hence it is easy to manage.

Disadvantages of the waterfall model

- The amount of risk is high.
- This model is not good for complex and object oriented projects
- Not suitable for the projects where requirements are at a moderate to high risk of changing. So, risk and uncertainty is high with this process model.
- Poor model for long and ongoing projects.
- Cannot accommodate changing requirements.

4.1.2 V model

The major drawback of waterfall model is we move to the next stage only when the previous one is finished and there was no chance to go back if something is found wrong in later stages. V-Model provides means of testing of software at each stage in reverse manner.

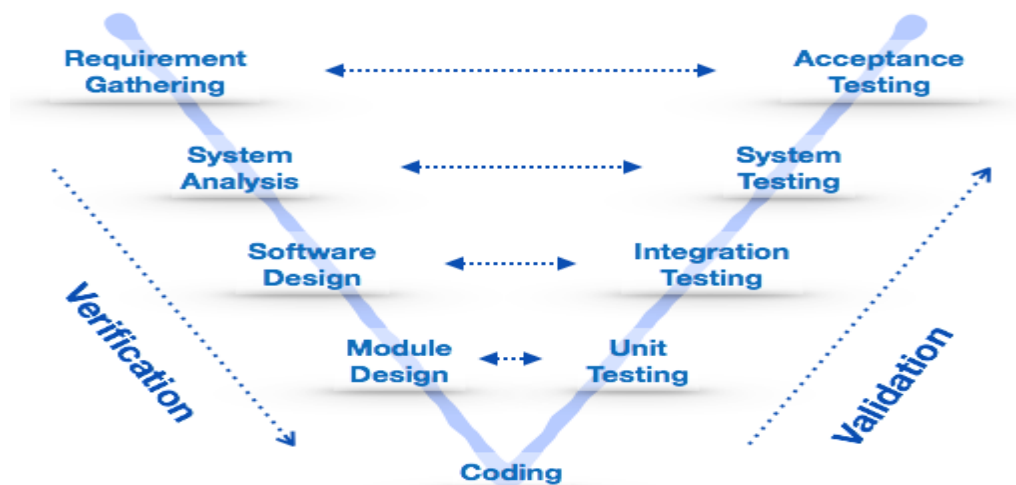


Fig 4.2

At every stage, test plans and test cases are created to verify and validate the product according to the requirement of that stage. For example, in requirement gathering stage the test team prepares all the test cases in correspondence to the requirements. Later, when the product is developed and is ready for testing, test cases of this stage verify the software against its validity towards requirements at this stage.

This makes both verification and validation go in parallel. This model is also known as verification and validation model.

When to use the V-Model?

- It is used in small and not so large projects where requirements are well defined and clear.
- It is used in projects where the acceptance criteria for the software are correctly formulated.
- It is used in projects where there are a lot of resources available having the required technical expertise

4.1.3 Incremental Process Models

The Incremental Process Model is a method of software development where the system is built step by step. Instead of delivering the whole system at once, it is developed and delivered in small parts called increments.

When an incremental model is used, the first increment is often a core product. That is, basic requirements are addressed but many supplementary features (some known, others unknown) remain undelivered. The core product is used by the customer (or undergoes detailed evaluation)

As a result of use and/or evaluation, a plan is developed for the next increment. The plan addresses the modification of the core product to better meet the needs of the customer and the delivery of additional features and functionality. This process is repeated following the delivery of each increment, until the complete product is produced.

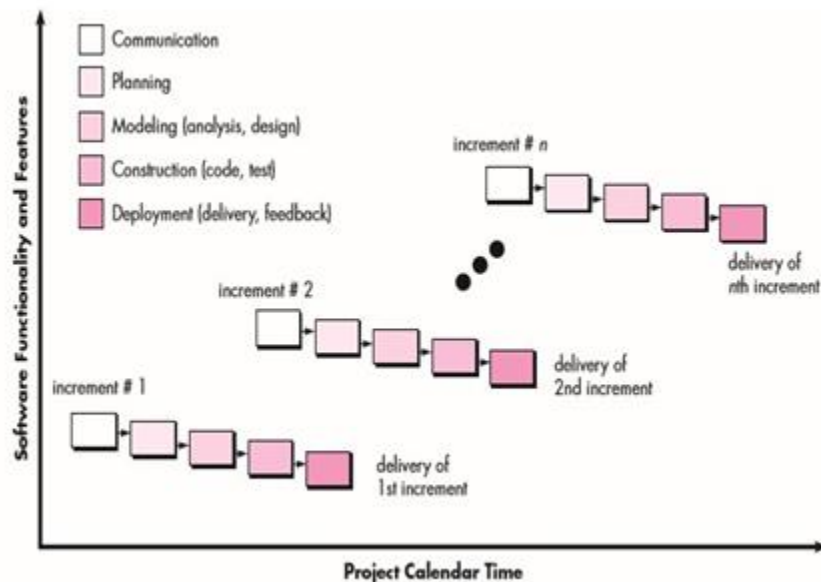


Fig 4.3: incremental model

Advantages of incremental model

- This model is flexible because the cost of development is low and initial product delivery is faster.
- It is easier to test and debug during the smaller iteration.
- The working software generates quickly and early during the software life cycle.
- The customers can respond to its functionalities after every increment

Disadvantages of the incremental model

- The cost of the final product may cross the cost estimate initially.
- This model requires a very clear and complete planning.
- The planning of design is required before the whole system is broken into small increments.
- The demands of customer for the additional functionalities after every increment causes problem during the system architecture.

4.1.3 Evolutionary Process Models

Evolutionary models are iterative type models. In the Iterative model, iterative process starts with a simple implementation of a small set of the software requirements and iteratively enhances the evolving versions until the complete system is implemented and ready to be deployed.

Following are the evolutionary process models.

- The prototyping model
- The spiral model

4.1.3.1 The Prototyping model: In the Prototyping Model, an initial prototype (a working version) of the software is built and refined through multiple iterations based on user feedback.

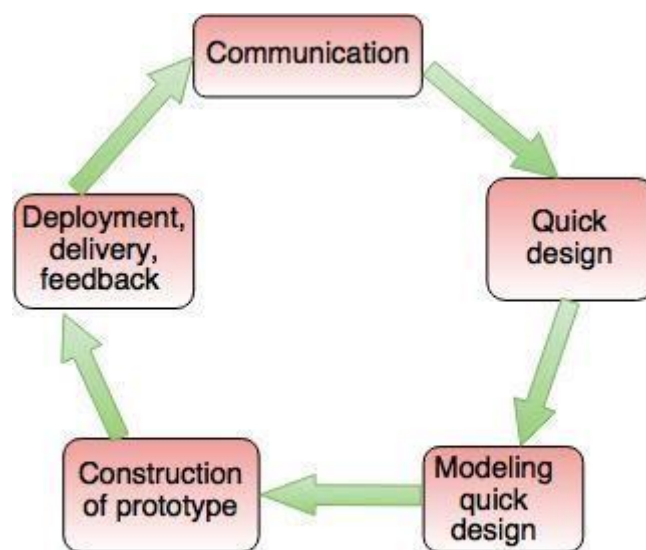


Fig. - The Prototyping Model

phases of Prototyping model

I. Communication

In this phase, developer and customer meet and discuss the overall objectives of the software.

II. Quick design

- Quick design is implemented when requirements are known.
- It includes only the important aspects like input and output format of the software.
- It focuses on those aspects which are visible to the user rather than the detailed plan.
- It helps to construct a prototype.

III. Modeling quick design

- This phase gives the clear idea about the development of software because the software is now built.
- It allows the developer to better understand the exact requirements.

IV. Construction of prototype

The prototype is evaluated by the customer itself.

V. Deployment, delivery, feedback

- If the user is not satisfied with current prototype then it refines according to the requirements of the user.
- The process of refining the prototype is repeated until all the requirements of users are met.
- When the users are satisfied with the developed prototype then the system is developed on the basis of final prototype.

Advantages of Prototyping Model

- Prototype model need not know the detailed input, output, processes, adaptability of operating system and full machine interaction.
- In the development process of this model users are actively involved.
- The development process is the best platform to understand the system by the user.
- Errors are detected much earlier.
- Gives quick user feedback for better solutions.
- It identifies the missing functionality easily. It also identifies the confusing or difficult functions.

Disadvantages of Prototyping Model:

- The client involvement is more and it is not always considered by the developer.
- It is a slow process because it takes more time for development.
- Many changes can disturb the rhythm of the development team.
- It is a thrown away prototype when the users are confused with it.

4.1.3.2 The Spiral model

Spiral model is a combination of both, iterative model and one of the SDLC model. It can be seen as if you choose one SDLC model and combine it with cyclic process (iterative model).

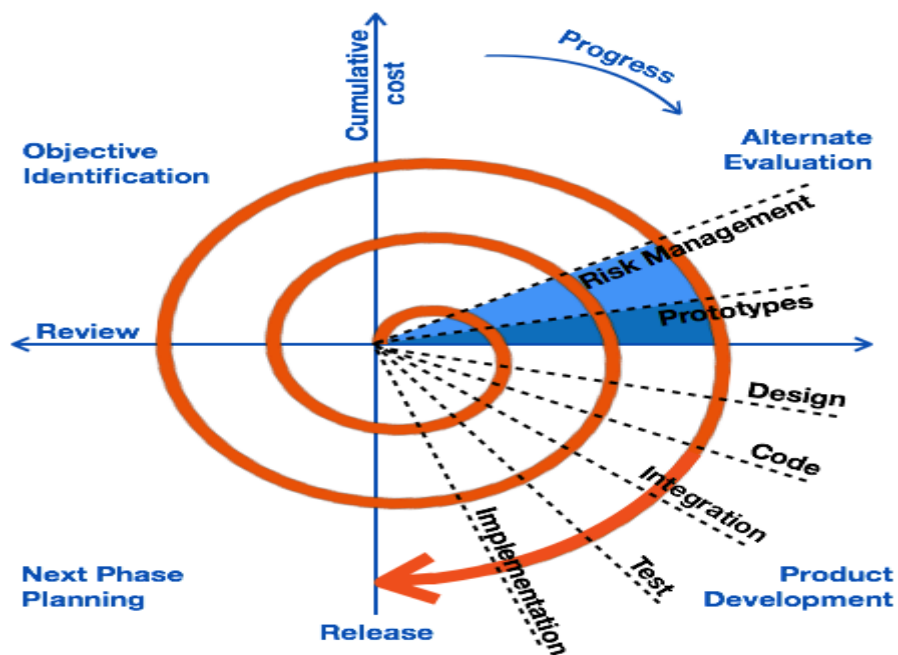


Fig 4.5: Spiral model

This model considers risk, which often goes un-noticed by most other models. The model starts with determining objectives and constraints of the software at the start of one iteration.

Next phase is of prototyping the software. This includes risk analysis. Then one standard SDLC model is used to build the software.

In the fourth phase of the plan of next iteration is prepared.

Advantages of Spiral Model

- It reduces high amount of risk.
- It is good for large and critical projects.
- It gives strong approval and documentation control.
- In spiral model, the software is produced early in the life cycle process.

Disadvantages of Spiral Model

- It can be costly to develop a software model.
- It is not used for small projects.

4.2 Comparison of Prescriptive Models

feature	Waterfall	V-Model	Prototyping	Incremental	Spiral
Flexibility	Low	Low	High	Medium	High
User Involvement	Low	Low	High	Medium	High
Risk Management	Low	Medium	Low	Medium	High
Best For	Small projects	Critical systems	UI-based software	Medium-large projects	Large, high-risk projects